STAT/MA 41600

In-Class Problem Set #31: October 28, 2026

1. Suppose that X and Y are continuous random variables whose joint probability density function $f_{X,Y}(x, y)$ is constant on the interior of the triangle with vertices at $(0, 0)$, $(8, 0)$, and $(0, 5)$, and otherwise $f_{X,Y}(x, y) = 0$.

1a. Find $P(Y > X)$.

1b. Find $P(X + Y \geq 3)$.

2. Suppose that $X_1, \ldots, X_n$ are n independent, continuous random variables, each of which is uniformly distributed on the interval $[0, 10]$. Let $Y = \max(X_1, \ldots, X_n)$.

2a. Find the cumulative distribution function of Y.

2b. Find the probability density function of Y.

3. Suppose that X has the cumulative distribution function:

$$F_X(x) = \begin{cases} 0 & \text{if } x \leq -20, \\ x/40 + 1/2 & \text{if } -20 < x < 20, \\ 1 & \text{if } x \geq 20. \end{cases}$$

3a. Is X a continuous uniformly distributed random variable? Why or why not?

3b. What is the probability density function of X?

3c. What is $E(X)$?

3d. What is $\mathrm{Var}(X)$?

4. Same setup as in Question **3**. In other words, please keep the same random variable X from Question **3**. Suppose that a and b are *positive*, constant real numbers, and define $Y = aX + b$.

4a. Is Y a continuous uniformly distributed random variable? Why or why not?

4b. What is the probability density function of Y?

4c. What is $E(Y)$?

4d. What is $\mathrm{Var}(Y)$?

5. Consider two independent continuous random variables X and Y.
The random variable X has a continuous uniform distribution on $[0, 2]$.
The pdf of Y is $f_Y(y) = 7e^{-7y}$ for $y > 0$, and otherwise $f_Y(y) = 0$.
Find $P(X > Y)$.